Evaluating Prompt Engineering Techniques for LLaMA-3: A Study of Zero-Shot, Few-Shot, and Chain-of-Thought Prompts Across Reasoning and Classification Tasks.

Darren Astle Travasso (P2841019@my365.dmu.ac.u)
Supervisor: Dr. Aboozar Taherkhani
De Montfort University, Leicester, UK

BACKGROUND:

Prompt engineering has emerged as a crucial approach for utilizing the potential of large language models (LLMs) without the necessity of elaborate fine-tuning. In contrast to traditional NLP systems that depend on labelled datasets and model retraining, prompt engineering allows users to obtain valuable outputs from LLMs by crafting effective prompts. This method has been particularly beneficial in scenarios with little or no resources, especially when domain-specific data is scarce (Brown, 2020).

Among the leading techniques for prompting are **zero-shot**, **few-shot**, and **chain-of-thought (CoT)** prompting. Zero-shot prompting involves asking the model to perform a task based solely on a natural language instruction, without any provided examples. This was illustrated with GPT-3, which showed unexpectedly strong results across various NLP tasks without the need for task-specific training (Brown, 2020). Incorporating a small number of task examples into the prompt through few-shot prompting boosts performance by guiding the model's predictions through imitation. This method has demonstrated enhancements in various tasks, including classification, summarization, and question answering. (Brown, 2020);(Min, 2022-10-20).

Chain-of-thought prompting is a newer development in which the prompt encourages the model to outline intermediate reasoning steps prior to providing a final answer. This technique has demonstrated significant improvements on complex reasoning tests, especially in the areas of mathematics and commonsense reasoning (Wei, 2022). It has been found effective even for models not explicitly trained for reasoning, provided the prompt includes step-by-step logical reasoning (Kojima, 2022).

These methods have been assessed using various datasets to examine their efficacy. For instance, **GSM8K** is widely utilized for solving arithmetic problems, **CommonsenseQA** emphasizes contextual and logical reasoning, and **SST-2** serves as a benchmark for sentiment classification. Each of these tasks assesses distinct abilities of LLMs, allowing for a thorough evaluation of the impact of prompt design on performance across reasoning and classification areas.(Wang, 2019-02-22); (Cobbe, 2021); (Talmor, 2019).

Prompt engineering, therefore, is not merely a technique but a critical interface between human instruction and model behaviour. As models grow more capable, the way tasks are framed through prompts becomes a determining factor in achieving optimal results.

INTRODUCTION:

The emergence of large language models (LLMs) like GPT-3, LLaMA, and others has greatly improved natural language comprehension and generation. These models have shown remarkable capabilities across a range of tasks, especially when provided with thoughtfully crafted prompts. Prompt engineering—the art of designing input prompts to achieve the best possible outputs from models—has become an effective and efficient substitute for traditional model fine-tuning. Nonetheless, the success of various prompting strategies depends on the specific task, data set, and the format of the prompt itself.

This dissertation examines the influence of three primary prompting methods—zero-shot, few-shot, and chain-of-thought—on the performance of LLMs across various NLP tasks. By implementing these techniques on three distinct datasets—GSM8K for mathematical reasoning, CommonsenseQA for

commonsense question answering, and SST-2 for sentiment analysis—I intend to assess the relative advantages and disadvantages of each prompting approach. The study further explores the significance of examples and reasoning prompts in improving model accuracy.

Through a series of organized experiments utilizing the LLaMA model, this research will shed light on how different prompting strategies affect performance, especially in low-resource or budget-limited settings where model fine-tuning is not an option. The findings of this study are anticipated to offer actionable insights for effective prompt creation and enhance the overall understanding of prompt engineering in LLMs.

PROPOSED WORK:

Aim

The primary aim of this project is to evaluate the effectiveness of various prompt engineering strategies on large language models (LLMs) across different natural language processing (NLP) task types, with a focus on classification and reasoning. The goal is to determine how prompt design choices—such as the number of in-context examples and the inclusion of reasoning steps—influence model performance.

Objectives

To achieve the aim, the following objectives have been defined:

1. **Implement Zero-shot, Few-shot, and Chain-of-Thought prompting techniques** across selected NLP datasets to assess their individual and combined impact.
2. **Evaluate model performance on reasoning and classification tasks** using GSM8K, CommonsenseQA, and SST-2 datasets respectively.
3. **Measure and compare performance metrics** including accuracy, F1 score, and solve rate for each prompt strategy.
4. **Analyze the influence of prompt structure and context size** on the model's ability to generalize across task types.
5. **Visualize performance trends** through graphical tools such as cumulative accuracy curves and metric comparison charts.
6. **Draw conclusions about the effectiveness of prompt engineering strategies** and identify which are best suited for specific task types.

Project Description

This project explores prompt engineering as an efficient and adaptable approach for regulating and enhancing the capabilities of large language models (LLMs) without necessitating extensive fine-tuning. Specifically, it examines the impact of various prompting techniques on model performance in reasoning and classification tasks. The methods under consideration include Zero-shot prompting (which lacks in-context examples), Few-shot prompting (that includes representative examples), and Chain-of-Thought prompting (which promotes sequential reasoning). Additionally, the research investigates the effect of combining Few-shot and Chain-of-Thought prompting to determine if integrating both illustrative examples and methodical reasoning yields beneficial results.

The selected experimental model for this study is the Meta LLaMA 3 8B Instruct, a leading open-weight instruction-tuned LLM. It is implemented with 4-bit quantization through the Hugging Face Transformers library, allowing for efficient inference on constrained computational resources while maintaining performance (Dettmers, 2022). The assessment covers three separate NLP datasets: GSM8K, which focuses on mathematical reasoning, CommonsenseQA for commonsense multiple-choice reasoning, and SST-2 for sentiment classification.

Each prompting approach is methodically applied to the datasets to evaluate model performance across varied task categories. Performance is quantified using standard metrics like accuracy, F1 score, and solve rate. Alongside the evaluation of individual strategies, a combination method—Few-shot with Chain-of-Thought—is utilized to determine if supplying multiple examples with detailed reasoning enhances the model's reasoning accuracy and generalizability beyond the capabilities of each individual method.

The results from these tests are analysed in both quantitative and visual formats, employing cumulative accuracy graphs and metric comparison charts to highlight trends. Through this extensive analysis, the project aims to offer deeper insights into the comparative advantages of different prompting techniques, particularly concerning their adaptability to specific task requirements. These results are anticipated to aid in the creation of more effective prompt engineering strategies for practical NLP applications.

RESEARCH QUESTIONS:

This research project aims to answer the following key questions:

1. **How does the number of in-context examples (zero vs. few) influence the performance of large language models across classification and reasoning tasks?**
   This question explores whether LLMs benefit significantly from few-shot prompting, and whether performance gains are consistent across tasks with varying complexity.
2. **What impact does the inclusion of reasoning structure (i.e. Chain-of-Thought prompting) have on the model's ability to solve tasks requiring logical or commonsense reasoning?**
   This assesses the extent to which explicit step-by-step reasoning enhances model performance on tasks like mathematical problem-solving and commonsense question answering.
3. **Does combining Few-shot prompting with Chain-of-Thought reasoning lead to performance gains beyond what is achieved through either strategy individually?**
   This evaluates whether the synergistic use of examples and structured reasoning improves task performance across domains.
4. **Are certain prompting strategies more effective for specific task types (e.g. classification vs. reasoning), and what factors contribute to these differences?**
   This question investigates whether the effectiveness of prompting strategies is task-dependent and what characteristics (e.g. format, reasoning complexity) influence these outcomes.

PROJECT REQUIREMENTS:

To execute the proposed research effectively and within the given timeframe, the following tools, platforms, and resources are necessary:

- **High-Performance GPU Access**: To meet the computational demands of large language models such as **Meta-Llama-3-8B-Instruct**, having access to high-performance GPU infrastructure is crucial. A subscription to **Google Colab Pro** has been obtained to benefit from improved GPU access (e.g., NVIDIA T4 or A100), quicker processing times, and longer session durations. This allows for seamless execution of model inference, dataset assessment, and data visualization.
- **Libraries and Frameworks**: The experimental workflow relies on robust and well-supported open-source libraries:
  - **Hugging Face Transformers** and **Datasets** for accessing pre-trained models and standardized datasets.
  - **PyTorch** for model inference and training tasks.
  - **Scikit-learn** for performance metrics such as accuracy, F1 score, and classification evaluation.
  - **Matplotlib** and **NumPy** for data visualization and numerical processing.
- **Datasets**:
  - **GSM8K**: To evaluate mathematical and step-by-step reasoning capabilities.

  - **CommonsenseQA**: For testing performance on commonsense and logical inference tasks.
  - **SST-2 (Stanford Sentiment Treebank)**: For binary sentiment classification.
- **Model Quantization Tools**: The **BitsAndBytes** library is used to apply 4-bit quantization, allowing for efficient model loading and inference even on limited-resource environments like Colab notebooks.
- **Reference Management Software**: **Zotero** is used for organizing and managing academic references in accordance with the **Harvard referencing style**, ensuring that all citations are correctly formatted and accessible throughout the writing process.
- **Documentation and Writing Tools**: Every written output, such as notes, logs, reports, and the final dissertation, is created using either **Microsoft Word** or **Overleaf (LaTeX)**, based on formatting needs and individual choice. These applications provide organized writing spaces and facilitate effective formatting, particularly for academic submissions.
- **Storage and Memory**: Google Colab Pro's extended memory and disk access are used for handling large datasets, caching model checkpoints, and storing intermediate outputs and result logs.

These tools and resources collectively support the efficient execution of experiments, documentation of findings, and preparation of high-quality, well-referenced academic outputs.

PROJECT RISKS:

This dissertation involves evaluating prompt engineering techniques (Zero-shot, Few-shot, and Chain-of-Thought) across multiple NLP tasks using datasets such as CommonsenseQA, GSM8K, and SST-2. Several risks have been identified that may impact project execution:

6.1 Technical Risks

- **Compute Limitations:** The project depends on open-source models that are run through Google Colab. Issues such as runtime limits, restricted GPU availability, or disconnections could interfere with the experiments. ***Mitigation:*** Tasks will be batched efficiently, and outputs saved incrementally to prevent data loss.
- **Model Variability and Evaluation Challenges:** Outputs from language models can be unpredictable, particularly in generative tasks. Assessing answers that rely on reasoning (such as GSM8K) introduces additional complexity. ***Mitigation:*** Seeds will be fixed where supported, and scoring will combine automated metrics with manual checks on a sample basis.

6.2 Methodological Risks

- **Prompt Generalization Across Tasks:** Prompt strategies might not yield consistent results across datasets that have varying task structures, such as classification compared to reasoning. ***Mitigation:*** Results will be analysed within the context of each dataset, avoiding overgeneralization.
- **Time Constraints for Experimentation:** Covering three datasets and multiple techniques within the project timeline poses a risk of incomplete experimentation. ***Mitigation:*** A clear execution plan prioritizes baseline results first, with additional experiments only if time allows.

6.3 Logistical and Personal Risks

- Data Loss or Session Interruptions: Working in cloud environments increases the risk of losing progress due to session timeouts. ***Mitigation:*** Regular backups and version control will be maintained throughout.
- Personal Schedule Conflicts: Overlaps with academic responsibilities or personal events may reduce available working time. ***Mitigation:*** A detailed timeline with built-in buffers ensures key milestones are met.

By identifying these key risks early and implementing practical mitigation strategies, the project is positioned for successful and professional execution within the allocated timeframe.

PROJECT SCHEDULE:

The dissertation project is organized into four structured phases, spanning from March to the end of May 2025. Each phase targets a specific set of objectives to ensure systematic progress from foundational research to final submission.

Phase 1 – Background Research and Proposal Development
Timeline: March – Mid-April

- Conduct an in-depth review of recent literature on prompt engineering in NLP.
- Identify and justify the core prompting strategies to be investigated: Zero-shot, Few-shot, and Chain-of-Thought.
- Select and evaluate suitable benchmark datasets (SST-2, CommonsenseQA, GSM8K) aligned with different NLP task types.
- Formulate research questions and define the experimental methodology.
- Draft, refine, and prepare the dissertation proposal for submission by **16th May 2025**.

Phase 2 – Prompt Design and Dataset Preparation
Timeline: Mid-April – Late April

- Develop and structure prompt templates for each selected technique (Zero-shot, Few-shot, and Few-shot with Chain-of-Thought).
- Format and preprocess all datasets to align with the input requirements of the selected LLM.
- Conduct small-scale pilot runs to validate prompt structure, evaluate model behaviour, and ensure compatibility with evaluation methods.

Phase 3 – Experimental Implementation
Timeline: Late April – Mid May

- Execute full-scale experiments across all three datasets, applying each prompting technique systematically.
- Log model outputs and compute relevant evaluation metrics (e.g., accuracy, F1-score, and qualitative reasoning assessments where applicable).
- Identify and address any emerging issues related to model limitations, dataset inconsistencies, or prompt sensitivity.

Phase 4 – Analysis and Dissertation Writing
Timeline: Mid-May – 30th May

- Analyze experimental results, comparing the performance of different prompting strategies across classification, reasoning, and QA tasks.
- Produce visualizations and tables to support critical discussion and interpretation of findings.
- Draft all chapters of the dissertation, including literature review, methodology, experimental results, and conclusions.
- Finalize and submit the complete dissertation document, along with associated code and supplementary materials, by **30th May 2025, 12:00 midday**.

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